INVESTIGATION REPORT

INV-2022-0045

NEXUS CK MB Calibration Drift Investigation

Investigation ID:	INV-2022-0045
Product:	NEXUS CK MB (Reference 30421)
Status:	Closed
Date Opened:	2022-09-15
Date Closed:	2022-11-20
Lead Investigator:	J. Martin - Quality Engineering
Approved By:	P. Dubois - Quality Director
Classification:	Minor - No product recall required

1. Executive Summary

Investigation INV-2022-0045 was initiated following observation of a gradual upward trend in S1 calibration signal for NEXUS CK MB lots manufactured between August and September 2022. Three consecutive lots (1009900820, 1009900821, 1009900822) showed S1 values above 530 RFV, approaching the upper specification limit of 550 RFV. No lots exceeded specifications, but the trend warranted investigation per SOP-QA-010.

The investigation determined that the root cause was gradual wear of the optical sensor on calibration station EQ-CAL-001, resulting in a systematic positive bias in fluorescence readings. The issue was resolved through preventive maintenance (sensor replacement) and enhanced trend monitoring was implemented.

2. Problem Description (QQOCQPC)

Qui (Who):

QC laboratory team (analysts QC-002, QC-003) performing routine calibration verification on NEXUS CK MB production lots.

Quoi (What):

Gradual upward drift in S1 calibration signal values. S1 readings increased from typical 500 RFV to 530-540 RFV over a 6-week period.

Ou (Where):

Calibration station EQ-CAL-001 at Marcy-l'Etoile production facility, QC laboratory area B.

Comment (How):

Trend identified during routine SPC chart review. S1 values showed consistent upward trend across consecutive lots while S2 remained stable.

Quand (When):

Trend first observed: August 2022. Investigation opened: 2022-09-15 after third consecutive lot above 530 RFV.

Pourquoi (Why):

Equipment aging - optical sensor degradation causing increased fluorescence readings. Normal wear pattern, but PM interval was insufficient to prevent measurable drift.

Combien (How much):

3 lots affected (1009900820-1009900822). No OOS results. No product impact - all lots released within specification. Estimated cost: 2,500 EUR (investigation + PM acceleration).

3. Data Analysis

3.1 Lot Chronology

Lot Number	Mfg Date	S1 (RFV)	S2 (RFV)	Status	Notes
1009900815	2022-07-10	498.2	101.5	Released	Normal
1009900816	2022-07-18	502.5	99.8	Released	Normal

Lot Number	Mfg Date	S1 (RFV)	S2 (RFV)	Status	Notes
1009900817	2022-07-25	505.1	100.2	Released	Normal
1009900818	2022-08-02	515.3	101.0	Released	Slight increase
1009900819	2022-08-10	522.8	100.5	Released	Trend noted
1009900820	2022-08-18	531.2	99.7	Released	Above 530 RFV
1009900821	2022-08-25	535.6	101.3	Released	Trend continues
1009900822	2022-09-02	538.9	100.8	Released	Investigation opened
1009900823	2022-09-15	501.3	100.1	Released	Post-PM (sensor replaced)

3.2 Equipment Analysis

Calibration station EQ-CAL-001 was inspected on 2022-09-12. The optical sensor showed measurable degradation in sensitivity, causing a systematic positive bias in fluorescence readings. S2 signal was unaffected because the S2 channel uses a separate optical path. Sensor was replaced during emergency PM (WO-2022-0310) on 2022-09-14.

3.3 Material Review

All SPR cone lots and reagent strips used during the affected period were reviewed. No material changes were identified. The same SPR lot (SPR-2022-04-001) was used throughout, ruling out material variation as a contributing factor.

4. Root Cause Analysis

Root Cause: Equipment aging - gradual degradation of the optical sensor on calibration station EQ-CAL-001 (S1 channel).

5M Category: Machine (Milieu/Equipment)

Contributing Factors:
- PM interval of 6 months was too long for the optical sensor component.
- SPC monitoring detected the trend, but alert threshold was set at specification limits rather than at a tighter warning level.

5. Conclusions and Corrective Actions

Conclusion: The calibration drift was caused by optical sensor degradation on EQ-CAL-001. No product was released outside specification. The issue was fully resolved by sensor replacement.

CAPA ID	Description	Type	Due Date	Status
CAPA-2022-0088	Reduce PM interval for optical sensors from 6 to 3 months	Preventive	2022-10-01	Completed
CAPA-2022-0089	Implement SPC warning limits at +/- 2 sigma (in addition to specification limits)	Preventive	2022-10-15	Completed
CAPA-2022-0090	Add S1 trend monitoring to monthly quality review dashboard	Preventive	2022-11-01	Completed

6. Effectiveness Verification

Post-CAPA monitoring (November 2022 - January 2023) confirmed that S1 signal values returned to nominal range (495-510 RFV) and remained stable. No recurrence of upward trend observed. Quarterly PM successfully executed on 2022-12-01 with no findings. CAPAs verified effective and closed on 2023-01-15.